



2D Kinematic analysis of walk, trot, and canter gaits in Criollo breed horses

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ABSTRACT: This study described the linear, temporal, and angular kinematic characteristics of thirty-five Criollo horses in walk, trot, and canter using 2D video motion capture and analysis. Retroreflective markers were placed on anatomical landmarks of the fore- and hindlimbs, and kinematic variables assessed with a motion analysis software. In walk, a mean stride length of 1.61 ± 0.14 m, stride duration of 0.90 ± 0.04 s, and mean velocity of 1.79 ± 0.13 m/s were observed. In trot, the mean stride length was 2.30 ± 0.23 m, stride duration was 0.60 ± 0.03 s, and the mean velocity was 3.82 ± 0.43 m/s. In canter, the stride length, stride duration, and velocity values were measured for the lead (2.86 ± 0.29 m, 0.51 ± 0.03 s, and 5.61 ± 0.64 m/s) and trail limbs (2.68 ± 0.25 m, 0.51 ± 0.02 s, and 5.28 ± 0.56 m/s). Regarding the angular variables, during the walk, the Criollo horses reached maximum protraction and retraction angles of the forelimbs of $16.90 \pm 2.31^\circ$ and $22.99 \pm 3.47^\circ$, respectively. For the hindlimbs, the maximum protraction was $10.86 \pm 2.45^\circ$, while the retraction was $31.47 \pm 2.13^\circ$. The joints numerically showed smaller ranges of motion (ROM) during the canter, especially in the shoulder, elbow, and hip joints. The values increased for walk and trot and decreased again for the canter, indicating a relationship between gait velocity and joint ROM. These results provided reference kinematic values for Criollo horses; further studies are needed to evaluate specific training or management applications.

Key words: equine biomechanics, gaits, musculoskeletal system, orthopaedics, kinesiology.

Análise cinemática 2D do passo, trote e canter de cavalos da raça Crioula

RESUMO: O objetivo principal deste estudo foi descrever as características cinemáticas lineares, temporais e angulares dos membros torácicos e posteriores de cavalos Crioulos no passo, trote e galope utilizando captura e análise de movimento em vídeo 2D. Trinta e cinco cavalos Crioulos clinicamente saudáveis (5 - 10 anos) foram filmados em condições controladas com um sistema de vídeo 2D a 240 fps. Marcadores retroreflexivos foram posicionados em pontos anatômicos dos membros torácico e pélvico, e as variáveis cinemáticas foram extraídas utilizando o software Quintic Biomechanics® v33. No passo, foram observados comprimento médio da passada de $1,61 \pm 0,14$ m, duração da passada de $0,90 \pm 0,04$ s e velocidade média de $1,79 \pm 0,13$ m/s. No trote, o comprimento médio da passada foi de $2,30 \pm 0,23$ m, a duração da passada foi de $0,60 \pm 0,03$ s e a velocidade média foi de $3,82 \pm 0,43$ m/s. No galope, os valores de comprimento da passada, duração da passada e velocidade para a guia ($2,86 \pm 0,29$ m, $0,51 \pm 0,03$ s e $5,61 \pm 0,64$ m/s) e membros da trilha ($2,68 \pm 0,25$ m, $0,51 \pm 0,02$ s e $5,28 \pm 0,56$ m/s). Em relação às variáveis angulares, durante o passo, os cavalos Crioulos atingiram ângulos máximos de protração e retração dos membros torácicos de $16,90 \pm 2,31^\circ$ e $22,99 \pm 3,47^\circ$, respectivamente. Para os membros posteriores, a protração máxima foi de $10,86 \pm 2,45^\circ$, enquanto a retração foi de $31,47 \pm 2,13^\circ$. As articulações apresentaram numericamente menores amplitudes de movimento (ADM) durante o galope, principalmente nas articulações do ombro, cotovelo e quadril. Os valores aumentaram para passo e trote e diminuíram novamente para galope, indicando relação entre velocidade de marcha e ADM articular. Estes resultados fornecem valores cinemáticos de referência para cavalos Crioulos; estudos adicionais são necessários para avaliar aplicações específicas em treinamento ou manejo.

Palavras-chave: biomecânica equina, andaduras, sistema musculoesquelético, clínica ortopédica, cinesiologia.

INTRODUCTION

The equine species encompasses a wide variety of breeds and biotypes, with conformational, morphometric, and biomechanical characteristics generally associated with breed selection standards (BEEMAN, 2008). In this context, the Criollo breed emerged as a central figure in Latin American equestrian sports by the late 20th century, solidifying

its role as a cultural and economic pillar across southern Brazil, Argentina, Chile, and Uruguay. This breed is deeply embedded in local culture due to the rural lifestyle heritage in these regions (KARLAU et al., 2023; VINOCUR et al., 2003). The rising popularity of the breed has driven the evolution of competitions aiming to reward horses that best combine morphological and functional traits within the category.

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The Criollo horse excels in both competitive equestrian events, such as the Freio de Ouro, where its agility and balance are demonstrated in manoeuvres like the *esbarrada* and *volta sobre patas*, and in practical fieldwork, particularly cattle handling, showcasing the breed's versatility and genetic refinement for both sport and ranch tasks across South America (PIZZI et al., 2024). The Freio de Ouro, the primary event for the morpho-functional selection of Criollo horses, is a rigorous competition held in Brazil that tests and highlights these animals' physical and psychological preparation, muscular strength, and endurance (CUCCO et al., 2016). This prestigious event serves as both a competitive showcase and a breeding evaluation, where only the finest horses advance in the breed registry, contributing to the selective improvement of Criollo lines. In 2022, genealogical records from the Associação Brasileira dos Criadores de Cavalos da Raça Crioula (ABCCC) – the main registry association for the breed – reported a remarkable 39.42% increase in registered animals, surpassing 460,000, the majority of which are concentrated in southern Brazil.

Conformational and performance traits are hereditary and influenced by environmental factors. Although the conformation of the Criollo breed has been studied by various authors, primarily based on body measurements and indices (DALTO, 2006; PIMENTEL et al., 2018a; PIMENTAL et al., 2018b), comprehensive studies on the gaits performed by this breed and their morphological influences are lacking. Although, linear and angular measurements of the limbs and joints, whether assessed statically or during motion, can help identify the types of movements horses perform, a complete 2D kinematic analysis can provide a quantitative analysis of each gait. Computerised kinematic analyses allow the examiner to record and assess the movements of the individual during one or more stride cycles (SANDBERG et al., 2020). This data can describe the geometry of movement in terms of displacement, velocity, and acceleration, facilitating the evaluation of joint angles, angular velocities, temporal data, and stride length (SUTHERLAND, 2002).

Numerous horse breeds have had their gait patterns characterized, but a comprehensive study has never been conducted for the Criollo breed (ROBILLIARD et al., 2007). Despite the breed's prevalence in South America, it remains relatively unknown within the global veterinary scientific community, particularly in terms of biomechanical studies. Previous studies have addressed the morphometric and biomechanical aspects of Criollo

horses (PAGANELA et al., 2012; PIMENTEL et al., 2018a; PIMENTEL et al., 2018b; VERNA et al., 2022a); however, they are scarce and do not encompass a complete kinematic analysis of the primary gaits these equines perform at a competitive level. Within this context, a series of investigations into the kinematic parameters of the breed was established to gather extensive information on all the different gaits performed by these individuals. Therefore, the primary objective of this study described the quantitative kinematic variables – linear, temporal, and angular – of the thoracic and pelvic limbs of Criollo horses in walk, trot, and canter.

MATERIALS AND METHODS

Study design

All procedures performed were approved by the Animal Ethics Committee (CEUA) of the Universidade Federal de Pelotas (UFPel), under registration number 51839-2019. The processing and analysis of the collected videos were conducted at the School of Equine and Veterinary Physiotherapy, part of Anglia Ruskin University, in the United Kingdom.

Experimental design

Thirty-five Criollo breed horses (*Equus caballus*) were evaluated, duly registered with the ABCCC. The horses were aged between 5 and 10 years, comprising 23 males and 12 females, with a mean weight of 428.81 ± 24.09 kg and a mean withers height of 1.42 ± 0.02 m. These horses originated from 10 training centres in the micro-region of Pelotas, in southern Rio Grande do Sul, Brazil. All selected horses were trained competitors in the Freio de Ouro discipline. The discipline requires horses to perform specific gaits and manoeuvres that showcase their agility and skill in handling cattle, replicating the demands of fieldwork. The equestrian surfaces were carefully controlled to match the conditions of the Freio de Ouro qualifying stages. The training routine for these horses consisted of a minimum two-year period involving a weekly exercise regimen, lasting around one hour per day. This regimen alternated between 10-minute walks and trots, as well as 10-minute canter exercises, followed by cool-down walks to stabilize heart rates to physiological limits. Each training centre employed its own rider responsible for the training, following similar routines at least five times per week. The competition regulations permit animals of any age and gender, with an equal number of male and female horses qualifying for the final stage.

Data collection was carried out at equestrian training centres in June 2022. All animals were subjected to the same predefined experimental and environmental conditions. Prior to kinematic analysis, the horses underwent a specific clinical examination of the locomotor system to determine the health status of the sample population. An experienced clinician conducted the evaluation and determined a grade of 0 on the American Association of Equine Practitioners (AAEP) lameness scale, indicating no lameness in the tested animals. All individuals were considered sound, with no presence of lameness or other musculoskeletal disorders.

The biotype of the sample was homogeneous and consistent with the phenotypic standards of the species, as the horses from the equestrian training centres belong to the Criollo breed and meet the category's requirements. Gender was not considered a factor as there are no significant morphometric differences between genders in this breed (PIMENTEL et al., 2018a).

Kinematic data collection

The positioning of markers and the setup of the study field were based on the methodology outlined by TORRES-PÉREZ et al. (2017). Twenty-four semi-spherical retroreflective markers (30mm in

diameter) were positioned and affixed with double-sided tape by the same operator on the right and left sides of the animals (Figure 1). The anatomical locations included the prominent anatomical landmarks of each limb: spine of the scapula, greater tubercle of humerus, lateral humeral epicondyle, ulnar carpal bone, lateral metacarpal epicondyle, distal interphalangeal joint, for the forelimbs and ventral aspect of the tuber coxae, great trochanter of the femur, lateral femoral epicondyle, lateral tubercle of the talus, lateral metatarsal epicondyle, and distal interphalangeal joint for the pelvic limbs.

The study area measured 10 metres in length and 3 metres in width, marked by cones for easy identification. Lateral space allowed for deceleration and repositioning of the animals, maintaining constant velocity and gait pattern, preventing transitions. A high-velocity camera (240fps, resolution 1280x550) was levelled horizontally on a fixed tripod 1 metre in height, positioned 10 metres from the centre of the platform. A 72W LED light was placed above the camera to activate the reflectivity of the markers. A 1-metre ruler was placed at the centre of the field, both horizontally and vertically, for system calibration. This setup was standardized for all training centres and repeated uniformly at each location. For this study, the horses were ridden. To guarantee the

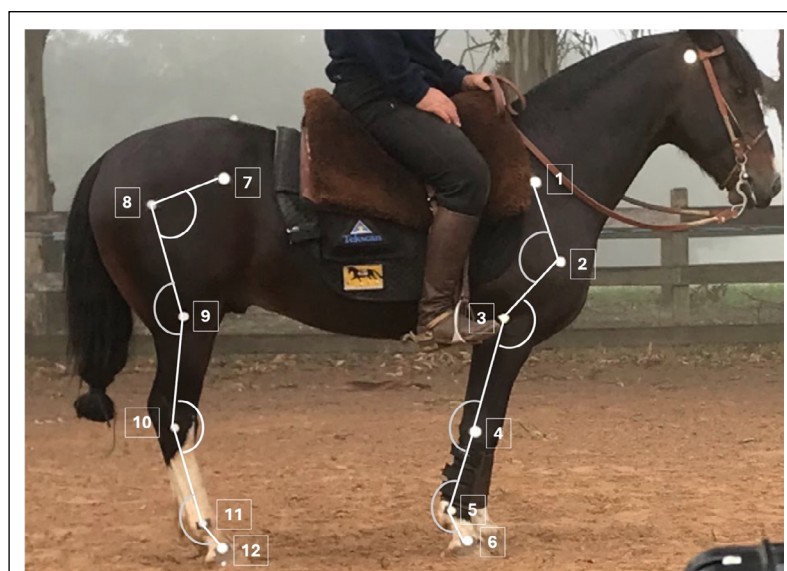


Figure 1 - Placement of retroreflective markers on thoracic and pelvic limbs of Criollo horses for kinematic analysis and angle measured during the analysis (1) spine of the scapula, (2) greater tubercle of humerus, (3) lateral humeral epicondyle, (4) ulnar carpal bone, (5) lateral metacarpal epicondyle, (6) distal interphalangeal joint, (7) ventral aspect of the tuber coxae, (8) Great trochanter of the femur, (9) Lateral femoral epicondyle, (10) Lateral tubercle of the talus, (11) Lateral metatarsal epicondyle, (12) distal interphalangeal joint.

accuracy of the kinematic analysis, each horse was equipped with its own tack, which was positioned to minimise any potential interference with the placement of the retroreflective markers. These markers were meticulously positioned to prevent any obstruction from the equipment, thereby ensuring precise measurement of the horses' movements.

Prior to data collection, the horses underwent a 10-minute warm-up at walk and trot. Subsequently, riders guided the animals in a straight line to the centre of the study field. Three slow-motion videos, each 10 seconds long, were recorded for each gait (walk, trot, and canter) from both sides for each horse, simulating the gaits performed during official competition trials. To ensure consistency in velocity across trials, each horse was conducted in a straight line to the centre of the study field, where they executed the gaits under controlled conditions that closely mimicked competition-level execution, thereby allowing for a reliable assessment of kinematic variables. After collection, the videos were processed and analyzed using the Quintic Biomechanics® v33 2D motion analysis system, where the obtained variables were processed and quantified.

Kinematic variables analysis

The analyzed variables were divided into temporal, linear, and angular categories and were measured individually for the thoracic and pelvic limbs from videos of both sides. Specifically, in the canter, it was also analyzed whether the filmed side was the lead or trail limb, based on the last thoracic limb to touch the ground during a complete stride cycle. Initially, all videos were measured in terms of stride length and duration, based on the first frame in which the thoracic limb touches the ground until the contact occurs again, completing the cycle of a full stride. Following this, the stride length-to-duration ratio was calculated to determine the velocity of each gait. Specifically, velocity was computed by dividing the measured stride length (the distance covered in one complete stride) by the corresponding duration (the time taken to complete that stride). Subsequently, within the group of temporal variables, the stance time(s), suspension time(s), and duty factor (examined as the ratio of stance time to the total stride time) were examined.

Finally, the angular variables investigated were measured following the methodology described by MORALES et al. (1998) and included the protraction angle (°), retraction angle (°), and the maximum fetlock extension angle during mid-stance (°), which is the angle where the respective limb had the fetlock perpendicular to the camera at the

intermediate point of the stance phase. Additionally, a tracking protocol was implemented, involving real-time angular mapping of the thoracic and pelvic limb joints throughout a complete stride cycle. This yielded minimum angles (flexion), maximum angles (extension), and ROM (range of motion), defined as the difference between the two, for the shoulder, elbow, carpus, forelimb fetlock, and proximal interphalangeal joints (thoracic limbs), and the hip, stifle, hock, hindlimb fetlock, and proximal interphalangeal joints (pelvic limbs). All analyses considered frames where the animal was closest to the centre of the study field to avoid perspective and parallax errors.

Statistical analysis

The three gaits were described according to the quantitative values of each kinematic variable. For this purpose, the mean values from the three collected videos of each side in walk, trot, and canter were considered; and subsequently, the mean value between the right and left sides was adopted. Descriptive statistical analyses were performed using IBM® SPSS® v29. Descriptive values of the final means and standard deviations (SD) of all linear, temporal, and angular kinematic variables were obtained for each thoracic (right: RF; left: LF) and pelvic limb (right: RH; left: LH).

RESULTS

A total of 18 videos were processed for each horse across three distinct gaits, equalling to a total of 630 strides per limb and 3,150 strides analysed. Gaits have been identified and classified according to ROBILLIARD et al. (2007).

Linear kinematic characteristics

The walk achieved a mean stride length of 1.61 ± 0.14 m, stride duration of 0.90 ± 0.04 s, and velocity of 1.79 ± 0.13 m/s. For the trot, the values found were: stride length 2.30 ± 0.23 m, stride duration 0.60 ± 0.03 s, and velocity 3.82 ± 0.43 m/s. In the canter, 54.44% of the videos showed animals executing a right lead canter (with the right forelimb being the last thoracic limb to touch the ground), and the stride length, stride duration, and velocity were measured for the lead limbs (2.86 ± 0.29 m, 0.51 ± 0.03 s, and 5.61 ± 0.64 m/s) and the trail limbs (2.68 ± 0.25 m, 0.51 ± 0.02 s, and 5.28 ± 0.56 m/s).

Temporal kinematic variables

The stance time of the thoracic limbs of Criollo horses during walking was 0.49 ± 0.03 s, and

for the pelvic limbs, it was 0.50 ± 0.04 s, while the suspension time was 0.35 ± 0.03 s and 0.33 ± 0.03 s for the respective limbs. The mean duty factor for the thoracic limbs was 0.54 ± 0.02 and 0.55 ± 0.03 for the pelvic limbs. In the trot, these values were 0.22 ± 0.03 s for both thoracic and pelvic limbs for stance time, and the swing time was 0.32 ± 0.03 s for the thoracic limbs and 0.33 ± 0.02 s for the pelvic limbs, with a mean duty factor of 0.35 ± 0.03 (thoracic) and 0.35 ± 0.04 (pelvic). Regarding the canter, both the lead thoracic and pelvic limbs showed a stance time of 0.15 ± 0.02 s, while the trail limb showed a stance time of 0.14 ± 0.02 s. The suspension time of the lead limbs (thoracic and pelvic) in this gait was 0.29 ± 0.03 s and 0.29 ± 0.02 , respectively. The values for this variable were 0.29 ± 0.02 s for the trail forelimb and 0.30 ± 0.03 s for the trail hindlimb. Finally, in the canter, the mean duty factor for the lead thoracic limb was 0.29 ± 0.03 , for the trail limb 0.28 ± 0.03 , for the lead pelvic limb 0.30 ± 0.04 , and for the trail pelvic limb 0.28 ± 0.04 . Stance time was the variable that showed the greatest numerical variation between gaits compared to suspension time, with both decreasing as velocity increased. The detailed values of the temporal variables by limb in each gait are presented in table 1.

Angular kinematic variables

In terms of angular variables, during the walk, Criollo horses reached a maximum protraction angle of $16.90 \pm 2.31^\circ$ and a retraction angle of $22.99 \pm 3.47^\circ$ for the thoracic limbs, with a mean hyperextension angle of $228.22 \pm 5.48^\circ$ at the fetlock during mid-stance. For the pelvic limbs, the maximum

protraction was $10.86 \pm 2.45^\circ$, while the retraction was $31.47 \pm 2.13^\circ$, and the hyperextension angle at the fetlock was $214.63 \pm 5.77^\circ$. Regarding the trot, the protraction angles for the thoracic limbs were $16.26 \pm 3.10^\circ$ and the retraction angles were $20.84 \pm 2.77^\circ$, with a maximum hyperextension angle at the fetlock of $239.6 \pm 6.40^\circ$ during stance. For the pelvic limbs, the maximum protraction and retraction angles, as well as the hyperextension angle at the fetlock during mid-stance, were $10.35 \pm 2.04^\circ$, $30.12 \pm 1.91^\circ$, and $234.88 \pm 9.98^\circ$, respectively. Table 2 shows detailed results of the angular variables for each thoracic and pelvic limb during the walk and trot, as well as the discriminated values for lead and trail limbs during the canter. The protraction and retraction angles were numerically similar between the thoracic and pelvic limbs with the change in velocity from walk to trot. These variables' values were higher in the canter for the lead limb, with greater protraction in the thoracic limbs and lower in the pelvic limbs compared to the trail limb, while the retraction angle was greater in both ipsilateral lead limbs. The mean trail angle increased numerically with the increase in velocity and with the change in gait in all limbs.

The shoulder joint of Criollo horses showed the smallest ROM in both the walk and trot, with mean values of $5.71 \pm 2.11^\circ$ and $6.25 \pm 3.30^\circ$, respectively. At the elbow joint, the amplitude was greater in the walk, with an angle of $48.33 \pm 6.36^\circ$, slightly decreasing in the trot to $46.60 \pm 9.81^\circ$. In the carpal region, the ROM during the walk was $64.44 \pm 5.66^\circ$, increasing to $73.50 \pm 7.65^\circ$ during the trot. The range of motion at the thoracic fetlock also showed an increase from the walk ($62.49 \pm 9.93^\circ$) to the trot

Table 1 - Means $\pm$ standard deviation of the temporal kinematic variables support time, suspension time and duty factor performed by Criollo breed horses (n = 35) in walk, trot and canter.

Variable	Gait	Limb							
		LFL	RFL	LHL	RHL	Lead FL	Trail FL	Lead HL	Trail HL
Stance time (s)	Walk	0.49 ± 0.02	0.49 ± 0.04	0.49 ± 0.04	0.51 ± 0.03				
	Trot	0.21 ± 0.03	0.22 ± 0.02	0.21 ± 0.03	0.22 ± 0.03				
	Canter					0.15 ± 0.02	0.14 ± 0.02	0.15 ± 0.02	0.14 ± 0.02
Suspension time (s)	Walk	0.35 ± 0.03	0.35 ± 0.03	0.33 ± 0.03	0.33 ± 0.03				
	Trot	0.32 ± 0.03	0.32 ± 0.02	0.33 ± 0.02	0.32 ± 0.02				
	Canter					0.29 ± 0.03	0.29 ± 0.02	0.29 ± 0.02	0.30 ± 0.03
Duty factor	Walk	0.53 ± 0.02	0.54 ± 0.02	0.54 ± 0.03	0.55 ± 0.02				
	Trot	0.34 ± 0.03	0.35 ± 0.03	0.34 ± 0.04	0.35 ± 0.04				
	Canter					0.29 ± 0.03	0.28 ± 0.03	0.30 ± 0.04	0.28 ± 0.04

LFL: left forelimb; RFL: right forelimb; LHL: left hindlimb; RHL: right hindlimb; Lead FL: lead forelimb; Trail FL: trail forelimb; Lead HL: lead hindlimb; Trail HL: trail hindlimb.

Table 2 - Means $\pm$ standard deviations of the angular kinematic variables of the protraction, retraction and fetlock middle-stance angles performed by Criollo breed horses (n = 35) in walk, trot and canter.

Angles (°)	Gait	Membro							
		LFL	RFL	LHL	RHL	Lead FL	Trail FL	Lead HL	Trail HL
Protraction (°)	Walk	17.05 $\pm$ 1.89	16.75 $\pm$ 2.73	10.94 $\pm$ 2.58	10.78 $\pm$ 2.32				
	Trot	16.02 $\pm$ 2.84	16.49 $\pm$ 3.35	9.89 $\pm$ 2.11	10.81 $\pm$ 1.96				
	Canter					28.61 $\pm$ 3.52	22.90 $\pm$ 3.08	18.69 $\pm$ 3.37	7.95 $\pm$ 2.35
Retraction (°)	Walk	22.60 $\pm$ 3.24	23.37 $\pm$ 3.70	31.47 $\pm$ 2.31	31.46 $\pm$ 1.94				
	Trot	20.71 $\pm$ 2.46	20.96 $\pm$ 3.08	30.55 $\pm$ 1.69	29.69 $\pm$ 2.12				
	Canter					18.15 $\pm$ 3.46	26.55 $\pm$ 3.08	30.45 $\pm$ 2.81	36.39 $\pm$ 2.56
Fetlock angle at middle-stance phase (°)	Walk	229.34 $\pm$ 5.82	227.10 $\pm$ 5.14	214.98 $\pm$ 5.66	214.27 $\pm$ 5.87				
	Trot	240.75 $\pm$ 6.01	238.45 $\pm$ 6.80	237.24 $\pm$ 6.38	232.51 $\pm$ 13.58				
	Canter					246.52 $\pm$ 7.35	244.81 $\pm$ 7.33	240.22 $\pm$ 7.85	241.53 $\pm$ 7.20

LFL: left forelimb; RFL: right forelimb; LHL: left hindlimb; RHL: right hindlimb; Lead FL: lead forelimb; Trail FL: trail forelimb; Lead HL: lead hindlimb; Trail HL: trail hindlimb.

(66.85 $\pm$ 10.70°). Regarding the pelvic limbs, the hip joint had a ROM of 16.71 $\pm$ 3.46° during the walk, decreasing to 14.45 $\pm$ 5.66 in the trot. The mean ROM values for the stifle joint were 26.24 $\pm$ 4.07° for the walk and 25.47 $\pm$ 4.64° for the trot, showing similar numbers in both gaits. In the tarsal region, individuals showed greater amplitude, with a ROM of 36.14 $\pm$ 6.01° during the walk, increasing to 42.19 $\pm$ 7.15° in the trot. However, at the level of the pelvic fetlock, the highest amplitude values were observed in both the walk and trot, being 72.46 $\pm$ 9.10° and 79.22 $\pm$ 8.69°, respectively. The ROM of all joints in the thoracic and pelvic segments showed smaller amplitudes during the canter for both lead and trail limbs. The maximum extension and flexion angles of the thoracic and pelvic limbs (walk and trot) and lead and trail limbs in the canter, as well as the ROM for each joint, are shown in table 3 and table 4.

DISCUSSION

This study provided a comprehensive kinematic analysis of the Criollo horse breed, focusing on key gait characteristics at walk, trot, and canter. The results established reference ranges for stride length, stride duration, and angular variables that distinguish the Criollo gait and offer a robust foundation for

further biomechanical studies. These findings serve as a baseline for assessing gait conformity to breed standards and understanding the breed's unique adaptations to various equestrian tasks, particularly those requiring endurance and agility.

The kinematic parameters observed in the Criollo horses reveal stride and angular characteristics that correspond well with their functional morphology, favouring sustained movement over high-speed exertion. This study identified marked differences in stride length and duration correlated with increasing velocity, in line with previous findings suggesting that higher velocities demand longer stride lengths and shorter stride durations to optimise locomotion efficiency (BACK & CLAYTON, 2013). Criollo horses' morphological traits – medium to short stature (1.40 to 1.55 m), compact muscular bodies, and robust limbs – are naturally adapted for endurance-oriented gaits rather than rapid bursts, contrasting with breeds bred for explosive or sustained high-speed events, such as the Quarter Horse and the Thoroughbred (VERGARA-HERNANDEZ et al., 2022). Specifically, Quarter Horses exhibit a shorter, muscular build well-suited for high-speed, short-duration events, while Thoroughbreds, with their leaner and taller physiques, are tailored for extended high-velocity performance. Criollo horses'

Table 3 - Means $\pm$ standard deviation of the flexion and extension angles ($^{\circ}$) and range of motion (ROM) ($^{\circ}$) of the thoracic limbs of competing Criollo breed horses (n = 35) at walk, trot and canter.

Joint	Gait	Limb-----											
		-----LFL-----			-----RFL-----			-----Lead FL-----			-----Trail FL-----		
		Flexion ($^{\circ}$)	Extension ($^{\circ}$)	ROM ($^{\circ}$)	Flexion ($^{\circ}$)	Extension ($^{\circ}$)	ROM ($^{\circ}$)	Flexion ($^{\circ}$)	Extensio n ($^{\circ}$)	ROM ($^{\circ}$)	Flexion ($^{\circ}$)	Extensio n ($^{\circ}$)	ROM ($^{\circ}$)
Shoulder	Walk	113.59 $\pm$ 8.21	119.86 $\pm$ 8.48	6.27 $\pm$ 2.50	114.28 $\pm$ 8.67	119.43 $\pm$ 9.07	5.15 $\pm$ 1.72						
	Trot	115.05 $\pm$ 8.21	121.27 $\pm$ 9.31	6.22 $\pm$ 3.24	114.14 $\pm$ 9.42	120.42 $\pm$ 9.93	6.28 $\pm$ 3.36						
	Canter							117.52 $\pm$ 10.31	122.71 $\pm$ 10.89	5.20 $\pm$ 3.86	115.91 $\pm$ 9.39	119.58 $\pm$ 9.74	3.68 $\pm$ 2.28
Elbow	Walk	105.78 $\pm$ 11.57	152.09 $\pm$ 10.33	46.31 $\pm$ 7.72	103.10 $\pm$ 8.19	151.43 $\pm$ 7.71	48.33 $\pm$ 4.99						
	Trot	104.05 $\pm$ 11.48	149.58 $\pm$ 10.88	45.53 $\pm$ 10.46	101.69 $\pm$ 9.93	149.35 $\pm$ 9.94	47.66 $\pm$ 9.15						
	Canter							112.28 $\pm$ 14.62	142.81 $\pm$ 9.96	30.53 $\pm$ 10.91	118.26 $\pm$ 12.48	147.65 $\pm$ 9.75	29.38 $\pm$ 8.27
Carpus	Walk	120.11 $\pm$ 7.09	184.66 $\pm$ 4.05	64.55 $\pm$ 6.27	119.21 $\pm$ 4.73	183.53 $\pm$ 2.69	64.32 $\pm$ 5.06						
	Trot	110.10 $\pm$ 7.44	182.54 $\pm$ 4.88	72.44 $\pm$ 7.81	109.05 $\pm$ 6.30	183.61 $\pm$ 4.81	74.55 $\pm$ 7.48						
	Canter							120.29 $\pm$ 9.28	162.70 $\pm$ 12.18	42.40 $\pm$ 9.97	123.02 $\pm$ 7.28	165.37 $\pm$ 9.11	42.35 $\pm$ 8.74
Fetlock	Walk	167.09 $\pm$ 9.69	231.72 $\pm$ 10.19	64.62 $\pm$ 8.03	167.67 $\pm$ 12.29	228.01 $\pm$ 11.58	60.35 $\pm$ 11.82						
	Trot	162.52 $\pm$ 11.03	230.65 $\pm$ 12.78	68.13 $\pm$ 10.93	162.51 $\pm$ 8.38	228.08 $\pm$ 11.44	65.57 $\pm$ 10.46						
	Canter							157.08 $\pm$ 13.54	200.93 $\pm$ 16.39	43.86 $\pm$ 12.93	161.09 $\pm$ 10.67	204.11 $\pm$ 13.91	43.02 $\pm$ 13.26

LFL: left forelimb; RFL: right forelimb; Lead FL: lead forelimb; Trail FL: trail forelimb.

gait parameters thus reflect a pattern that prioritises endurance, agility, and versatility, aligning well with the demands of equestrian competitions like the Freio de Ouro, where balance between stamina and agility is essential.

Furthermore, the stride characteristics documented in this study provide an effective baseline for breed-specific comparisons. For example, Quarter Horses and Thoroughbreds have been shown to achieve stride lengths of 5.90 ± 0.11 m and 4.40 ± 0.11 m, respectively, at gallop (VERGARA-HERNANDEZ et al., 2022), whereas Criollo horses demonstrated stride values more consistent with slower, sustained gaits. Rather than comparing to gallop data – since canter and gallop are biomechanically different gaits – this study focuses on the kinematic values observed in Criollo horses. Our results showed a mean stride length of 1.61 ± 0.14 m at the walk, which rises to 2.30 ± 0.23 m at the trot and further to 2.86 ± 0.29 m at the canter. These increasing values reflect the expected progression of stride extension with gait speed and

provide specific reference values for future intra-breed or matched-gait comparisons. As expected, velocity increased proportionally with each gait transition (e.g., from 1.79 ± 0.13 m/s at walk to 5.61 ± 0.64 m/s at canter), which corresponds with similar kinematic changes documented in other breeds (CLAYTON, 1994a; CLAYTON, 1994b).

Recent literature underscores that as horses' speed increases, so too do their stride frequency and limb angulation, with biomechanical adaptations to accommodate joint loading and muscle activation demands (CLARK et al., 2020; PUGLIESE et al., 2020). These adaptations reflect compensatory responses necessary for supporting efficient motion at higher velocities, as evidenced in racing breeds (BARREY, 2013). However, it is worth noting that while Criollo horses' kinematic profiles reveal characteristics suited to their endurance-based activities, extrapolations regarding injury risk or specific training protocols should be treated cautiously, as they exceed the scope of this study's findings.

Table 4 - Means $\pm$ standard deviation of the flexion and extension angles ($^{\circ}$) and range of motion (ROM) ($^{\circ}$) of the pelvic limbs of competing Criollo breed horses (n = 35) at walk, trot and canter.

Joint	Gait	Limb-----											
		LHL			RHL			Lead HL			Trail HL		
		Flexion ($^{\circ}$)	Extension ($^{\circ}$)	ROM ($^{\circ}$)	Flexion ($^{\circ}$)	Extension ($^{\circ}$)	ROM ($^{\circ}$)	Flexion ($^{\circ}$)	Extension ($^{\circ}$)	ROM ($^{\circ}$)	Flexion ($^{\circ}$)	Extension ($^{\circ}$)	ROM ($^{\circ}$)
Hip	Walk	88.86 $\pm$ 7.84	104.96 $\pm$ 8.79	16.10 $\pm$ 2.87	88.78 $\pm$ 7.15	106.09 $\pm$ 8.17	17.32 $\pm$ 4.04						
	Trot	90.49 $\pm$ 8.49	106.28 $\pm$ 11.83	15.78 $\pm$ 7.79	92.72 $\pm$ 7.77	105.85 $\pm$ 7.54	13.12 $\pm$ 3.52						
	Canter							90.83 $\pm$ 10.38	99.35 $\pm$ 8.50	8.52 $\pm$ 5.32	97.25 $\pm$ 8.14	107.19 $\pm$ 7.17	9.93 $\pm$ 4.03
Stifle	Walk	134.72 $\pm$ 7.78	160.44 $\pm$ 7.27	25.72 $\pm$ 3.99	134.48 $\pm$ 7.42	161.23 $\pm$ 6.40	26.75 $\pm$ 4.15						
	Trot	125.06 $\pm$ 7.38	151.02 $\pm$ 7.34	25.96 $\pm$ 3.94	127.46 $\pm$ 8.76	152.44 $\pm$ 7.53	24.98 $\pm$ 5.35						
	Canter							131.89 $\pm$ 7.43	146.81 $\pm$ 7.39	14.91 $\pm$ 4.62	131.69 $\pm$ 7.91	145.20 $\pm$ 7.74	13.50 $\pm$ 4.74
Hock	Walk	128.71 $\pm$ 8.69	165.04 $\pm$ 4.47	36.33 $\pm$ 6.86	129.68 $\pm$ 5.33	165.64 $\pm$ 4.40	35.96 $\pm$ 5.15						
	Trot	116.39 $\pm$ 9.15	160.77 $\pm$ 4.15	44.38 $\pm$ 7.04	121.51 $\pm$ 8.54	161.51 $\pm$ 3.94	40.00 $\pm$ 7.27						
	Canter							131.35 $\pm$ 8.66	158.00 $\pm$ 4.63	26.65 $\pm$ 7.84	132.82 $\pm$ 8.60	158.65 $\pm$ 4.69	25.84 $\pm$ 7.72
Hindlimb Fetlock	Walk	150.44 $\pm$ 10.91	223.93 $\pm$ 9.85	73.49 $\pm$ 9.80	150.00 $\pm$ 8.38	221.43 $\pm$ 6.67	71.42 $\pm$ 8.40						
	Trot	148.02 $\pm$ 8.84	229.99 $\pm$ 9.91	81.97 $\pm$ 8.46	149.86 $\pm$ 8.49	226.32 $\pm$ 8.39	76.46 $\pm$ 8.91						
	Canter							152.37 $\pm$ 9.93	201.33 $\pm$ 17.62	48.96 $\pm$ 16.62	147.48 $\pm$ 11.40	199.48 $\pm$ 15.97	52.00 $\pm$ 13.67

LHL: left hindlimb; RHL: right hindlimb; Lead HL: lead hindlimb; Trail HL: trail hindlimb.

Criollo horses, noted for their versatility across equestrian disciplines, demonstrated kinematic adaptability, particularly in trot and canter gaits, akin to those observed in dressage and reining horses. This versatility is a critical asset in the Freio de Ouro competition, which assesses manoeuvrability and comfort, especially in cattle tasks. The trot and canter velocities documented here (mean trot: 3.61 ± 0.10 m/s, canter: 3.91 ± 0.17 m/s) align with previous dressage studies (CLAYTON, 1994a; CLAYTON, 1994b), supporting the functional adaptability of the Criollo gait. However, breed classification relies on specific locomotion patterns, as noted by ROBILLIARD et al. (2007).

Across the three primary gaits – walk, trot, and canter – stance and swing times were similar, indicating balanced temporal dynamics with minimal asymmetry. In Criollo horses, the near-identical stance and swing times at walk, trot, and canter not only correspond with the gait-speed – dependent saddle pressure patterns that centralize load under the rider's saddle, but also exploit elastic

strain energy storage in distal limb tendons to reduce active muscular work by up to 80%, and align with the third metacarpal's nonuniform strain energy density distributions that protect bone integrity under varying load conditions (FRUEHWIRTH et al., 2004; HARRISON et al., 2010; RUBIN et al., 2013). Comparative studies have shown that while general temporal trends appear across breeds, Criollo-specific values reflect the breed's unique conformation and biomechanics (ROBILLIARD et al., 2007). Further research could explore how training interventions affect temporal parameters, with longitudinal studies offering insights into gait dynamics over time. By establishing these kinematic benchmarks, this study enhanced biomechanical understanding and practical applications in Criollo equestrian activities.

Comparison to other breeds

Comparison to published kinematic studies on other breeds underscores the specificity of Criollo's gait dynamics. In the absence of kinetic or metabolic measurements, purely kinematic

comparisons – based on stride length and duty factor (MARTINS et al., 2025; PIZZI et al., 2024; VERNA et al., 2022b) – underscore the unique, economical gait of Criollo horses across walk, trot and canter. In the current study, Criollo horses demonstrated stride characteristics indicative of an economical gait pattern that aligns with their use in endurance events, such as the Freio de Ouro. Such breed-specific kinematic benchmarks allow for meaningful comparisons with other breeds' gait parameters (BACK & CLAYTON, 2013). Future research could further enhance this comparative analysis by incorporating standardized measurements across multiple breeds under similar conditions to verify these kinematic distinctions.

Limitations of the study

Some limitations should be acknowledged to contextualize these findings. Firstly, the sample of horses included in this study may not fully represent the variability within the Criollo population, especially given the regional focus of the sample. Future studies involving a more diverse sampling of Criollo horses across broader geographical areas would further solidify these reference kinematic values.

Additionally, the assessment of soundness was based on a subjective evaluation rather than an objective lameness detection method. Implementing validated methods, such as inertial sensor-based lameness detection, could improve the accuracy of the initial health assessment and reduce potential biases from subjective observation. Another limitation involves the single-camera setup utilized, which restricts the analysis to a two-dimensional plane. As such, minor misalignments within the capture volume may have impacted the accuracy of certain kinematic variables. Employing multi-camera setups or three-dimensional kinematic analysis could yield a more comprehensive understanding of these gait patterns and minimise parallax errors.

Furthermore, it is worth noting that while this study captured and quantified gait parameters effectively, it did not regulate horse velocity precisely across all trials. Kinematic variables are sensitive to speed, and while every effort was made to maintain a consistent gait, slight velocity variations may have influenced certain parameters. Future studies may benefit from controlled speed trials to further refine the accuracy of the kinematic data obtained.

Future directions

This study established foundational kinematic benchmarks for Criollo horses, offering

key gait parameters essential for advancing biomechanical research and practical applications. Defining normal ranges for stride length, duty factor, and angular measurements, the study supports a detailed understanding of the breed's movement profile, which is invaluable given the Criollo's prominence in Brazilian equestrian activities alongside breeds like the Quarter Horse (MINISTÉRIO DA AGRICULTURA, 2016; DA SILVA & SILVA FARIAS, 2017). The kinematic data presented not only aid in characterising the Criollo gait but also underscore the need for breed- and discipline-specific research, crucial for investigating functional asymmetries and discerning pain-related gait changes (EGAN et al., 2019).

As technology advances, the ability to capture biomechanical data in natural, outdoor settings has become increasingly relevant, allowing researchers to simulate realistic training environments that complement laboratory findings (GALISTEO et al., 2010). Establishing kinematic values for Criollo horses represents an unprecedented contribution, supporting further studies on locomotor conditions, training optimisations, and performance assessments across diverse equestrian contexts. With emerging technologies enabling more sophisticated biomechanical and kinetic measurements, future work could integrate 3D analyses, metabolic cost assessments and longitudinal monitoring to deepen our understanding of Criollo performance and welfare. This study, however, establishes key 2D kinematic benchmarks for walk, trot and canter in clinically sound Criollo horses, providing a solid reference for intra-breed and longitudinal gait comparisons.

CONCLUSION

This study successfully characterized the kinematic variables of walk, trot, and canter gaits in Criollo horses, providing a crucial reference for future biomechanical research. Understanding these parameters is vital for optimising training, rehabilitation, and performance evaluation, benefiting both the scientific community and equestrian professionals.

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DECLARATION OF CONFLICT OF INTEREST

The authors declared that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

AUTHORS' CONTRIBUTIONS

All authors contributed equally for the conception and writing of the manuscript. All authors critically revised the manuscript and approved of the final version.

BIOETHICS AND BIOSECURITY COMMITTEE APPROVAL

All procedures performed were approved by the Animal Use Ethics Committee (CEUA) of the Universidade Federal de Pelotas (UFPe), under registration 51839-2019.

DATA AVAILABILITY STATEMENT

Not applicable.

DECLARATION OF USE OF ARTIFICIAL INTELLIGENCE

We declare that no artificial intelligence resources were used in the writing or development of this manuscript.

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